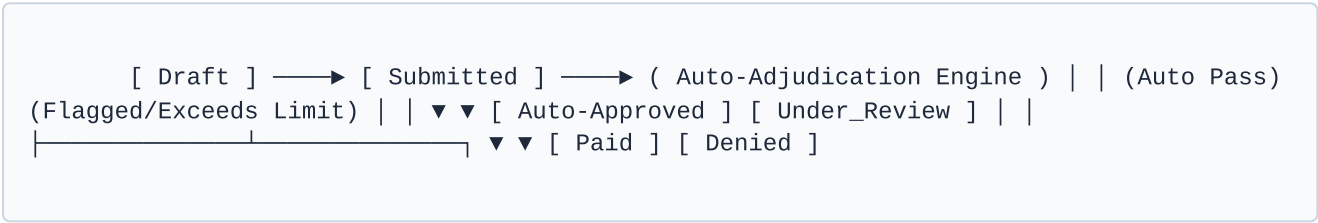
Claims Processing Logic Handbook

Standard Operating Procedures & Automated Logic Engine Specifications for Prototyping

1. OVERVIEW & OPERATIONAL FLOW

This handbook outlines the standard logic, validation schemas, business rules, and state machine transitions required to build a functional claims processing prototype. The system manages the entire claim lifecycle from initial submission (First Notice of Loss) through automated adjudication, manual audit, and payout authorization.



2. CORE DATA SCHEMA

The following entity definitions must be enforced across data input validation and persistent storage models.

Field Name	Data Type	Allowed Values / Constraints	Prototype Validation & Business Logic
`claim_id`	String	Format: `CLM-YYYY-XXXX`	Auto-generated unique primary key upon submission.
`claimant_id`	String	FK (User UUID)	Must map to an active, verified account in good standing.
`claim_type`	Enum	`Health`, `Travel`, `Expense`, `Auto`	Determines dynamically required attachment fields and approval chains.
`amount_claimed`	Decimal	`> 0.00`	Evaluated against rule matrix for auto-approval vs manual tiering.
`receipts_attached`	Array[URL]	File links	Mandatory array if `amount_claimed` exceeds $25.00.
`claim_status`	Enum	`Draft`, `Submitted`, `Under_Review`, `Approved`, `Denied`, `Paid`	State variable controlled strictly by state machine transition hooks.

3. BUSINESS RULES ENGINE MATRIX

All submitted claims must pass sequentially through the rule processing engine before state assignment.

- **Rule 1 (Duplicate Prevention):** If `claimant_id`, `amount_claimed`, and `incident_date` match an existing claim within 30 days → Flag status to `Under_Review` with system tag `FLAG_DUPLICATE`.
- **Rule 2 (Fast-Pass Auto-Approval):** If `amount_claimed` ≤ $100.00 AND policy age > 90 days AND `receipts_attached` count ≥ 1 → Transition directly to `Auto-Approved`.
- **Rule 3 (Tiered Escalation Matrix):**
  - Claims $0.01 – $500.00: Assignable to Level 1 Junior Adjuster.
  - Claims $500.01 – $5,000.00: Assignable to Level 2 Operations Manager.
  - Claims > $5,000.00: Requires Level 3 Executive Sign-off & Audit Log Entry.
- **Rule 4 (SLA & Timeout Deadlines):**
  - Intake & Initial System Validation: Target < 60 seconds.
  - Manual Adjudication Target SLA: 5 Business Days (escalates on Day 6).
  - Payout Execution: Trigger payment API within 3 business days post-approval.

> **Implementation Note for Developers:** Ensure state updates trigger appropriate webhook notifications to the claimant's registered endpoint or notification service.